

LibeRality reference manual

Model-Informed Optimal Clinical Trial Design

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32 help topics

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Launch the LibeRality optimal-design workbench

Description

Launch the LibeRality optimal-design workbench

Usage

```
liberality_gui(  
  design = NULL,  
  criterion = lity_criterion_D(),  
  queue = NULL,  
  host = "127.0.0.1",  
  port = NULL,  
  launch.browser = TRUE  
)
```

Arguments

design	Initial design; the teaching example is used when omitted.
criterion	Initial criterion.
queue	Optional LibeRties local or remote queue.
host, port, launch.browser	Passed to [shiny::runApp()]. Set 'launch.browser = NULL' to return the Shiny application object for hosted deployment.

Value

Invisibly, the Shiny app.

LibeRality React workbench widget

Description

LibeRality React workbench widget

Usage

```
liberality_workbench(  
  payload,  
  input_id = "liberality_workbench",  
  width = NULL,  
  height = "100vh",  
  elementId = NULL  
)
```

Arguments

payload	Workbench payload.
input_id	Shiny event prefix.
width, height	Widget dimensions.
elementId	Optional element id.

Shiny output for the LibeRality workbench

Description

Shiny output for the LibeRality workbench

Usage

```
liberalityWorkbenchOutput(outputId, width = "100%", height = "100vh")
```

Arguments

outputId	Output id.
width, height	CSS dimensions.

Define a trial arm or elementary design

Description

Define a trial arm or elementary design

Usage

```
lity_arm(  
  name,  
  events,  
  size = 1L,  
  allocation = 1,  
  population = "default",  
  costs = list(),  
  sample_volume = 0,  
  metadata = list()  
)
```

Arguments

name	Arm name.
events	Event records, commonly from [lity_schedule()].
size	Exact number of subjects assigned to the arm.
allocation	Approximate allocation weight.
population	Population/stratum name.
costs	Named fixed, per-subject, per-visit, per-sample, and assay costs.

sample_volume

Blood/sample volume per observation.

metadata

Additional metadata.

Define a design constraint

Description

Define a design constraint

Usage

```
lity_constraint(  
  name,  
  type = c("min_separation", "max_samples", "total_subjects",  
"total_cost",  
    "max_blood_volume", "max_duration", "arm_size", "allocation",  
"max_rse",  
    "minimum_power", "exposure", "custom"),  
  limit,  
  arm = NULL,  
  endpoint = NULL,  
  parameters = NULL,  
  lower = NULL,  
  upper = NULL,  
  function_value = NULL,  
  metadata = list()  
)
```

Arguments

name	Constraint name.
type	Constraint type.
limit	Upper or lower limit, depending on 'type'.
arm	Optional arm.

endpoint	Optional endpoint.
parameters	Optional parameter selection for precision constraints.
lower, upper	Optional interval bounds.
function_value	Custom function accepting a design and returning a scalar.
metadata	Additional metadata.

Evaluate all design constraints

Description

Evaluate all design constraints

Usage

```
lity_constraint_check(design, evaluation = NULL)
```

Arguments

design	Design object.
evaluation	Optional cached design evaluation.

Restore a LibeRality object from its semantic contract

Description

Remote JSON transport deliberately discards arbitrary R attributes. This function rebuilds LibeRality classes exclusively from validated 'liberality.*' schema identifiers, including classes nested inside results.

Usage

```
lity_contract_restore(x)
```

Arguments

x
A list decoded from a LibeRality wire result.

Value

The recursively reconstructed LibeRality object.

Define an optimal-design criterion

Description

Define an optimal-design criterion

Usage

```
lity_criterion(  
  type = .lity_criterion_types,  
  name = NULL,  
  parameters = NULL,  
  weights = NULL,  
  contrast = NULL,  
  matrix = NULL,  
  target = NULL,  
  threshold = NULL,  
  alpha = 0.05,  
  alternative = c("two.sided", "greater", "less"),  
  margin = 0,  
  effect = NULL,  
  base = NULL,  
  components = NULL,  
  direction = NULL,  
  scale = NULL,  
  reference = NULL,  
  nsim = 250L,  
  seed = 7301L,  
  utility = NULL,  
  metadata = list()  
)  
  
lity_criterion_D(parameters = NULL, ...)  
  
lity_criterion_A(parameters = NULL, weights = NULL, ...)  
  
lity_criterion_E(parameters = NULL, ...)  
  
lity_criterion_Ds(parameters, ...)  
  
lity_criterion_c(contrast, ...)
```

```
lity_criterion_L(matrix, weights = NULL, ...)

lity_criterion_rse(
  parameters = NULL,
  weights = NULL,
  summary = c("weighted", "max", "mean"),
  ...
)

lity_criterion_prediction(gradient, ...)

lity_criterion_bayesian(base = lity_criterion_D(), weights = NULL, ...)

lity_criterion_robust(base = lity_criterion_D(), weights = NULL, ...)

lity_criterion_minimax(base = lity_criterion_A(), ...)

lity_criterion_maximin(base = lity_criterion_D(), ...)

lity_criterion_model_average(base = lity_criterion_D(), weights = NULL,
...)

lity_criterion_precision_probability(parameters = NULL, threshold = 20,
...)

lity_criterion_discrimination(
  method = c("KL", "T", "model_discrimination"),
  weights = NULL,
  ...
)

lity_criterion_power(
  contrast,
  effect = NULL,
  alpha = 0.05,
  alternative = c("two.sided", "greater", "less"),
  margin = 0,
  kind = c("power", "superiority", "noninferiority"),
  ...
)

lity_criterion_target(target, nsim = 250L, seed = 7301L, ...)
```

```

lity_criterion_correct_dose(target, nsim = 250L, seed = 7301L, ...)

lity_criterion_expected_utility(
  target = NULL,
  utility = NULL,
  weights = NULL,
  nsim = 250L,
  seed = 7301L,
  ...
)

lity_criterion_cost(...)

lity_criterion_burden(weights = NULL, ...)

lity_criterion_compound(
  components,
  weights = NULL,
  scale = NULL,
  reference = NULL,
  ...
)

lity_criterion_pareto(components, ...)

```

Arguments

type	Criterion type.
name	Display name.
parameters	Parameter subset.
weights	Parameter, scenario, component, or loss weights.
contrast	Contrast/gradient vector.

matrix	L-criterion matrix.
target	Clinical, precision, or decision target.
threshold	Criterion threshold.
alpha	Type-I error for power criteria.
alternative	One- or two-sided alternative.
margin	Non-inferiority/superiority margin.
effect	Assumed effect; defaults to the contrast applied to nominal parameters.
base	Base criterion for Bayesian/robust aggregation.
components	Criteria combined into a compound or Pareto objective.
direction	Optimisation direction.
scale, reference	Compound-objective scaling.
nsim, seed	Simulation controls for decision criteria.
utility	Optional custom utility function; built-in serializable utility definitions are preferred for queue execution.
metadata	Additional metadata.

...	Additional arguments passed from a criterion-specific helper to [lity_criterion()].
summary	RSE aggregation: weighted, maximum, or arithmetic mean.
gradient	Parameter gradient for expected prediction variance.
method	Model-discrimination distance: KL, T, or weighted general model discrimination.
kind	Hypothesis criterion: power, superiority, or non-inferiority.

Define a complete optimal-design problem

Description

Define a complete optimal-design problem

Usage

```
lity_design(  
  model,  
  arms,  
  endpoints = NULL,  
  population = lity_population(),  
  scenarios = NULL,  
  alternative_models = NULL,  
  variables = list(),  
  constraints = list(),  
  prior_fim = NULL,  
  name = "Optimal design",  
  description = "",  
  metadata = list()  
)
```

Arguments

model	Primary LibeRation 'nm_model'.
arms	Named list of [lity_arm()] objects.
endpoints	Named list of [lity_endpoint()] objects.
population	Population strata.
scenarios	Parameter/model/operational scenarios.

alternative_models	Competing LibeRation models for discrimination.
variables	Optimisable variables.
constraints	Design constraints.
prior_fim	Optional prior Fisher information matrix.
name, description	Human-readable identity.
metadata	Additional serializable metadata.

Define an outcome used by an optimal design

Description

Define an outcome used by an optimal design

Usage

```
lity_endpoint(  
  name,  
  type = c("continuous", "binary", "ordinal", "count", "time_to_event",  
    "recurrent_event"),  
  dvid = 1L,  
  link = NULL,  
  scale = c("linear_predictor", "response"),  
  thresholds = NULL,  
  distribution = NULL,  
  dispersion = NULL,  
  target = NULL,  
  metadata = list()  
)
```

Arguments

name	Outcome name.
type	Continuous, binary, ordinal, count, time-to-event, or recurrent-event.
dvid	Dataset outcome identifier.
link	Link applied to model predictions for non-continuous outcomes.
scale	Whether model predictions are a linear predictor or already on the response scale.

thresholds

Ordered cut points for an ordinal outcome.

distribution

Distribution detail, such as 'poisson', 'negative_binomial', 'exponential', or 'weibull'.

dispersion

Optional count dispersion or Weibull shape.

target

Optional clinical target definition.

metadata

Additional serializable metadata.

Convert a LibeRator endpoint into a LibeRality design endpoint

Description

Convert a LibeRator endpoint into a LibeRality design endpoint

Usage

```
lity_endpoint_from_liberator(endpoint, dvid = 1L, type = "continuous")
```

Arguments

endpoint	A 'lator_endpoint'.
dvid	Outcome identifier.
type	Statistical response type used by the design model.

Evaluate one or more criteria for a design

Description

Evaluate one or more criteria for a design

Usage

```
lity_evaluate(  
    design,  
    criteria = lity_criterion_D(),  
    tolerance = 1e-10,  
    check_constraints = TRUE  
)
```

Arguments

design	Design object.
criteria	Criterion or named list of criteria.
tolerance	Numerical rank tolerance.
check_constraints	Evaluate constraints after criteria.

Value

A 'lity_evaluation'.

A complete teaching example

Description

Creates a two-arm oral PK design with sampling-time and allocation variables, parameter uncertainty, operational constraints, and representative criteria.

Usage

```
lity_example()
```

Value

A list containing model, design, and criteria.

Validate LibeRality against PopED and PFIM

Description

Runs matched population-FO designs through LibeRality, PopED and PFIM, transforms residual-error parameters to a common variance scale, and compares complete Fisher matrices, RSEs, determinants and runtimes.

Usage

```
lity_external_validate(  
  fixtures = lity_external_validation_fixtures(),  
  engines = c("PopED", "PFIM"),  
  repetitions = 3L,  
  tolerance = list(absolute = 1e-05, relative = 1e-04, rse = 0.01),  
  output_dir = NULL,  
  design_search = TRUE  
)  
  
## S3 method for class 'lity_external_validation'  
print(x, ...)
```

Arguments

fixtures	Fixture identifiers returned by <code>lity_external_validation_fixtures()</code> .
engines	External engines to run in addition to LibeRality.
repetitions	Number of warm-core and end-to-end timing repetitions.
tolerance	Named list containing absolute, relative and RSE tolerances. RSE is measured in percentage points.
output_dir	Optional output directory for RDS, CSV, JSON, matrix and HTML artifacts.

`design_search`

Whether to run the matched D-optimal candidate-grid search.

`x`

External-validation result.

`...`

Unused.

Value

An object of class `lity_external_validation`.

List external-validation fixtures

Description

Lists matched designs available for PopED and PFIM cross-validation.

Usage

```
lity_external_validation_fixtures()
```

Value

Character vector of versioned fixture identifiers.

Write an external-validation report

Description

Writes a self-contained HTML summary of numerical agreement, coverage and runtimes.

Usage

```
lity_external_validation_report(x, file)
```

Arguments

x	A <code>lity_external_validation</code> object.
---	---

file	Output HTML file.
------	-------------------

Value

Normalized report path.

Calculate expected information for a design

Description

Continuous outcomes use the multivariate-normal population FIM, including mean/covariance cross-information and exact LibeRation prediction sensitivities. Non-continuous outcomes use distribution-aware working moments, allowing shared random effects and variance-component information.

Usage

```
lity_information(  
  design,  
  scenario = 1L,  
  model = NULL,  
  tolerance = 1e-10,  
  approximation = c("full_gaussian", "fo_block")  
)
```

Arguments

design	LibeRality design.
scenario	Scenario name, index, or object.
model	Optional model override.
tolerance	Numerical rank tolerance.
approximation	Information approximation. <code>"full_gaussian"</code> includes covariance derivatives for fixed effects; <code>"fo_block"</code> uses the conventional block-diagonal population-FO form used by PopED and PFIM.

Value

A 'lity_information' object.

Create a typed LibeRties optimal-design job

Description

Create a typed LibeRties optimal-design job

Usage

```
lity_job(  
  design,  
  criterion = lity_criterion_D(),  
  operation = c("optimise", "evaluate", "simulate", "pareto"),  
  arguments = list(),  
  label = NULL  
)
```

Arguments

design	Serializable design.
criterion	Criterion.
operation	Evaluate, optimise, simulate, or Pareto exploration.
arguments	Additional operation arguments.
label	Job label.

Import a reviewed LibeRary model

Description

Import a reviewed LibeRary model

Usage

```
lity_model_from_liberary(library_id, root = NULL, allow_draft = FALSE)
```

Arguments

library_id	Catalogue identifier.
root	Optional catalogue root.
allow_draft	Permit a draft entry.

Summarise empirical operating characteristics

Description

Summarise empirical operating characteristics

Usage

```
lity_operating_characteristics(simulation, alpha = 0.05)
```

Arguments

simulation	A result from [lity_simulate_trials()].
------------	---

alpha	Confidence interval alpha.
-------	----------------------------

Optimise a LibeRality design

Description

Optimise a LibeRality design

Usage

```
lity_optimise(  
  design,  
  criterion = lity_criterion_D(),  
  method = c("auto", "L-BFGS-B", "Nelder-Mead", "pso",  
"coordinate_exchange", "hybrid",  
  "multiplicative", "fedorov_wynn"),  
  control = list(),  
  penalty = 1e+08,  
  seed = 7301L,  
  progress = NULL,  
  cancel = NULL,  
  checkpoint = NULL  
)
```

Arguments

design	Design with [lity_variable()] definitions.
criterion	Scalar criterion. Compound criteria are supported directly; use [lity_pareto()] for an unscalarised Pareto criterion.
method	Automatic, gradient-based, simplex, particle-swarm, coordinate exchange, hybrid, multiplicative, or Fedorov-Wynn optimisation.
control	Named optimiser controls.
penalty	Quadratic constraint-violation penalty.

seed

Reproducible seed.

progress

Optional callback receiving every evaluation record.

cancel

Optional function returning 'TRUE' to cancel.

checkpoint

Optional RDS checkpoint path.

Value

A 'lity_optimisation' result.

Explore the Pareto frontier of a multi-objective design

Description

Explore the Pareto frontier of a multi-objective design

Usage

```
lity_pareto(  
  design,  
  criterion,  
  n = 250L,  
  control = list(),  
  seed = 7301L,  
  progress = NULL  
)
```

Arguments

design	Design object.
criterion	Pareto criterion.
n	Number of candidate designs.
control	Optional controls; 'n' overrides the argument.
seed	Reproducible seed.
progress	Optional callback.

Define populations and covariate strata

Description

Define populations and covariate strata

Usage

```
lity_population(  
  strata = data.frame(name = "default", weight = 1),  
  name = "Population",  
  metadata = list()  
)
```

Arguments

strata	Data frame with a stratum 'name', 'weight', and optional model covariates.
--------	--

name	Population set name.
------	----------------------

metadata	Additional metadata.
----------	----------------------

Generate a reproducible optimal-design report

Description

Generate a reproducible optimal-design report

Usage

```
lity_report(  
  x,  
  file = tempfile("LibeRality-report-", fileext = ".html"),  
  title = NULL  
)
```

Arguments

x	Design, evaluation, optimisation, Pareto, or simulation result.
---	---

file	Output HTML path.
------	-------------------

title	Optional report title.
-------	------------------------

Value

Normalized report path, invisibly.

Define a parameter, model, or operational scenario

Description

Define a parameter, model, or operational scenario

Usage

```
lity_scenario(  
  name,  
  theta = NULL,  
  omega = NULL,  
  sigma = NULL,  
  model = NULL,  
  probability = 1,  
  covariates = list(),  
  dropout = 0,  
  adherence = 1,  
  missed_sample = 0,  
  metadata = list()  
)
```

Arguments

name	Scenario name.
theta, omega, sigma	Optional model parameter values.
model	Optional alternative LibeRation model.
probability	Scenario probability.
covariates	Named covariate overrides.

dropout, adherence, missed_sample

Operational probabilities.

metadata

Additional metadata.

Create a NONMEM-style trial schedule

Description

Create a NONMEM-style trial schedule

Usage

```
lity_schedule(  
  sampling_times,  
  dose = 0,  
  dose_times = 0,  
  dose_cmt = 1L,  
  observation_cmt = 1L,  
  rate = 0,  
  duration = NULL,  
  ii = 0,  
  addl = 0L,  
  ss = 0L,  
  route = "extravascular",  
  dvid = 1L,  
  covariates = list(),  
  id = 1L  
)
```

Arguments

sampling_times	Observation times, or a data frame containing 'TIME' and optional 'DVID' and 'CMT' columns.
dose	Dose amount(s).
dose_times	Dose administration times.
dose_cmt, observation_cmt	Dose and observation compartments.

rate	Infusion rate. Zero represents a bolus or extravascular dose.
duration	Optional infusion duration; converted to 'rate'.
ii, addl	NONMEM repeat-dose interval and additional-dose count.
ss	NONMEM steady-state flag.
route	Descriptive administration route.
dvid	Outcome identifier for vector 'sampling_times'.
covariates	Named constant covariates copied to every record.
id	Subject identifier used by the elementary design.

Value

A NONMEM-style event data frame with design metadata.

Simulate complete trials under a LibeRality design

Description

Simulate complete trials under a LibeRality design

Usage

```
lity_simulate_trials(  
  design,  
  n = 100L,  
  scenarios = design$scenarios,  
  fit = FALSE,  
  method = "FOCEI",  
  seed = 7301L,  
  n_cores = 1L,  
  retain_data = TRUE,  
  progress = NULL  
)
```

Arguments

design	Design object.
n	Number of replicated trials.
scenarios	Scenario sampling probabilities; defaults to the design.
fit	Whether to re-estimate each simulated trial with LibeRation.
method	Estimation method when 'fit = TRUE'.
seed	Reproducible seed.

n_cores	Simulation cores passed to LibeRation.
---------	--

retain_data	Retain simulated datasets.
-------------	----------------------------

progress	Optional callback.
----------	--------------------

Value

A 'lity_simulation' result.

Export a final design to LibeRation

Description

Export a final design to LibeRation

Usage

```
lity_to_liberation(  
  design,  
  simulate = FALSE,  
  seed = 7301L,  
  workspace = NULL,  
  project = NULL  
)
```

Arguments

design	Design or optimisation result.
--------	--------------------------------

simulate	Generate a stochastic dataset instead of event templates.
----------	---

seed	Simulation seed.
------	------------------

workspace	Optional LibeRation workspace.
-----------	--------------------------------

project	Optional project id/name. A project and first model version are created when both workspace and project are supplied.
---------	---

Value

Dataset or project metadata.

Validate a LibeRality design

Description

Validate a LibeRality design

Usage

```
lity_validate(design, strict = FALSE)
```

Arguments

<code>design</code>	Design object.
---------------------	----------------

<code>strict</code>	Whether warnings should be returned as validation failures.
---------------------	---

Value

Validation report.

Define an optimisable design variable

Description

Define an optimisable design variable

Usage

```
lity_variable(  
  name,  
  target = c("sampling_time", "dose", "rate", "duration", "arm_size",  
"allocation",  
    "covariate"),  
  arm,  
  index = 1L,  
  lower = -Inf,  
  upper = Inf,  
  values = NULL,  
  initial = NULL,  
  type = c("continuous", "integer", "discrete", "categorical"),  
  covariate = NULL,  
  metadata = list()  
)
```

Arguments

name	Variable name.
target	'sampling_time', 'dose', 'rate', 'duration', 'arm_size', 'allocation', or 'covariate'.
arm	Target arm name.
index	Observation/dose index when relevant.
lower, upper	Bounds.

values	Candidate values for discrete or categorical variables.
--------	---

initial	Initial value; inferred from the design when omitted.
---------	---

type	Continuous, integer, discrete, or categorical.
------	--

covariate	Covariate column for a covariate variable.
-----------	--

metadata	Additional metadata.
----------	----------------------

Execute a typed LibeRality worker task

Description

Execute a typed LibeRality worker task

Usage

```
lity_worker_task(model, design, arguments = list())
```

Arguments

model	Serialized primary model retained by the queue contract.
design	Serialized design.
arguments	Worker arguments.

Render a LibeRality workbench

Description

Render a LibeRality workbench

Usage

```
renderLiberalityWorkbench(expr, env = parent.frame(), quoted = FALSE)
```

Arguments

expr	Widget expression.
env	Evaluation environment.
quoted	Whether expression is quoted.
